11 Chemistry
Unit 2 Topic Test 5: Calorimetry

Details	
Student Name:	Teacher Name:
Date Set: ____/____/26	Due Date: ____/____/26

Curriculum Assessed

On completion of Unit 2 the student should be able to measure and analyse chemicals in water, including the energy changes of reactions measured in water. This topic test assesses the specific heat capacity of water, $q = mc\Delta T$, solution and combustion calorimetry with water, energy per gram and per mole, the percentage of energy transferred, and heat losses and other sources of error, with linked mole and gas ($pV = nRT$) calculations.

Task Overview

Section A – Multiple Choice: 10 questions, 1 mark each (10 marks). Answer on the separate multiple-choice answer sheet.
Section B – Short Answer: 4 questions (30 marks). Answer in this booklet.

Conditions

- Test conditions (revision topic test)
- VCE Chemistry Data Book provided
- Permitted materials: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator (no CAS).
- Not permitted: notes of any kind, blank paper, white-out, mobile phones, smart watches or any other electronic device.
- 46 mins writing time (about 1.15 minutes per mark)

Marking

- 40 marks
- Marked according to teacher-made marking guide.
- Show all working in calculation questions; give final answers with units and an appropriate number of significant figures. Equations must be balanced and include states.

A+	A	B+	B	C+	C	D+	D	E+	E	NS
90-100%	80-89%	75-79%	70-74%	65-69%	60-64%	55-59%	50-54%	45-49%	40-44%	<40%

Total Marks:		Percentage:		Letter Grade:	

SECTION A – Multiple-choice questions (10 marks)

Answer **all** questions on the separate multiple-choice answer sheet by filling in the letter of the best response. Each question is worth 1 mark. No marks are deducted for incorrect answers. Do **not** write your Section A answers in this booklet.

Question 1

When solid potassium chloride, KCl, dissolves in water in a polystyrene cup, the temperature of the water falls.

Which one of the following is correct for this process?

	Type of process	Direction of energy transfer
A.	exothermic	from the dissolving KCl to the water
B.	exothermic	from the water to the dissolving KCl
C.	endothermic	from the water to the dissolving KCl
D.	endothermic	from the dissolving KCl to the water

Question 2

Solution calorimetry experiments are often done in a polystyrene cup with a lid. The main reason is that

A. it reduces the transfer of energy between the solution and the surroundings.

B. polystyrene has a high specific heat capacity, so it stores the energy released.

C. polystyrene absorbs no energy at all from the solution.

D. polystyrene speeds up the reaction, so the maximum temperature is reached sooner.

Question 3

Four samples of water, H_2O, are heated. Which sample needs the most energy?

A. 100 g of water heated by 30 °C

B. 200 g of water heated by 20 °C

C. 50 g of water heated by 50 °C

D. 300 g of water heated by 8 °C

Question 4

The heat of combustion of octane, C_8H_{18}, is 47.9 kJ g^{-1}.

The energy released when 1.00 mol of octane burns completely is closest to

A. 0.420 kJ

B. 47.9 kJ

C. 5.46×10^6 kJ

D. 5.46×10^3 kJ

Question 5

A small solar water-heating panel absorbed 60.0 kJ of energy from sunlight while heating 500 g of water, H_2O. The water temperature increased by 14.0 °C.

The percentage of the absorbed solar energy that reached the water is closest to

A. 51.2%

B. 48.8%

C. 205%

D. 0.0488%

Question 6

The temperature of a solution in a polystyrene cup was recorded every 30 s. The reading at 30 s was taken immediately before a second solution was added and mixed in.

Time (s)	0	30	60	90	120	150	180
Temperature (°C)	19.6	19.6	25.3	27.4	27.8	27.5	27.1

Using the highest temperature recorded (without extrapolating), which value of ΔT should be used to calculate the energy released by the reaction?

A. 8.2 °C

B. 7.5 °C

C. 2.5 °C

D. 27.8 °C

Use the following information to answer Questions 7 and 8.

When anhydrous copper(II) sulfate, $CuSO_4$, dissolves in water, energy is released. A student dissolved 6.38 g of anhydrous $CuSO_4$ (0.0400 mol) in 50.0 g of water, H_2O, in a polystyrene cup. The water temperature increased by 12.2 °C. Assume that all the energy released is absorbed by the 50.0 g of water.

Question 7

The energy released per mole of $CuSO_4$ is closest to

A. 2.55 kJ mol^{-1}

B. 0.400 kJ mol^{-1}

C. 6.37×10^4 kJ mol^{-1}

D. 63.7 kJ mol^{-1}

Question 8

In a repeat of this experiment, the student forgot to stir, and some solid $CuSO_4$ was still undissolved when the highest temperature was recorded. The calculation still used 0.0400 mol of $CuSO_4$.

The calculated energy released per mole of $CuSO_4$ in the repeat would be

A. unchanged, because dividing by the amount of $CuSO_4$ cancels the effect.

B. higher, because undissolved solid absorbs extra energy from the water.

C. lower, because less energy was released, but the energy was still divided by 0.0400 mol.

D. unchanged, because the temperature rise does not depend on how much solid dissolves.

Question 9

Student 1 dissolved 2.00 g of a salt in 50.0 g of water and measured the temperature rise. Student 2 dissolved 4.00 g of the same salt in 100.0 g of water, starting at the same temperature, in the same type of cup. Ignore heat losses.

Compared with Student 1, which one of the following is correct for Student 2?

	Temperature rise	Calculated energy released per mole of salt
A.	the same	the same
B.	twice as large	the same
C.	the same	twice as large
D.	half as large	the same

Question 10

In two experiments, 1.00 g of fuel X (molar mass 30.0 g mol^{-1}) and 1.00 g of fuel Y (molar mass 58.0 g mol^{-1}) were each burned to heat 100 g of water in the same apparatus. The water temperature rose by 11.0 °C with X and by 9.0 °C with Y. Assume the same fraction of energy reached the water each time.

Which one of the following is correct?

	Fuel releasing more energy per gram	**Fuel releasing more energy per mole**
A.	X	X
B.	X	Y
C.	Y	X
D.	Y	Y

END OF SECTION A

SECTION B – Short-answer questions (30 marks)

Answer **all** questions in the spaces provided. Show all working in calculation questions. Give final answers with units and an appropriate number of significant figures. Equations must be balanced and include states, for example $H_2(g)$, NaCl(s).

Question 1 (8 marks)

Self-heating cans of coffee, sold for hikers and for emergency kits, have an inner compartment of calcium oxide, CaO (quicklime). Pressing a button on the base releases water onto the CaO, and the reaction releases energy that heats the coffee.

$$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(s)$$

To test the reaction, Tran added 4.20 g of CaO to 100.0 g of water, H_2O, in an insulated cup. The water temperature increased from 17.0 °C to 27.8 °C. Assume that all the energy released by the reaction was absorbed by the 100.0 g of water.

a) Calculate the amount, in mol, of CaO used.

1 mark

b) Calculate the energy, in J, absorbed by the water.

2 marks

c) Calculate the energy released per mole of CaO, in kJ mol^{-1}.

2 marks

d) A can must heat 200.0 g of coffee from 17.0 °C to 65.0 °C. Assume that the coffee has the same specific heat capacity as water and that all the energy released reaches the coffee. Using your answer to part c), calculate the minimum mass of CaO that the can must contain.

3 marks

Question 2 (9 marks)

Many Victorian homes have a gas storage hot-water service: natural gas, which is mainly methane, CH_4, burns beneath a tank of water, H_2O. To estimate how much of the energy from the gas reaches the water, a family turned off all their other gas appliances and recorded the gas meter while the tank heated.

The tank held 135 L of water (135 kg). The water was heated from 14.0 °C to 60.0 °C. The gas meter showed that 1.00 m^3 of methane, measured at 15.0 °C and 102 kPa, was burned.

Assume that the natural gas is pure methane and that it burns completely. The molar heat of combustion of methane is 890 kJ mol^{-1}.

a) Write a balanced equation for the complete combustion of methane, including states.

2 marks

b) Calculate the amount, in mol, of methane burned.

2 marks

c) Calculate the percentage of the energy released by the burning methane that was absorbed by the water.

3 marks

d) The final temperature was read from the tank's thermostat display, which measures the water at the top of the tank, where the water is hottest. Explain how using this reading as the final temperature of all the water affects the percentage calculated in part c).

2 marks

Question 3 (6 marks)

Jun investigated the energy released when lithium chloride, LiCl, dissolves in water. In each trial he dissolved 1.70 g of LiCl in 50.0 g of water, H_2O, in a polystyrene cup with a lid, stirred, and recorded the temperature once it had stopped rising. His results are shown below.

Trial	Initial temperature (°C)	Temperature recorded (°C)	Observation
1	20.4	27.6	all solid dissolved
2	20.6	27.6	all solid dissolved
3	20.5	24.1	temperature recorded 10 s after adding the LiCl, before it had stopped rising

a) Identify the trial whose result should not be used to calculate the energy released.

1 mark

b) Using the mean temperature change of the valid trials, and the mass of water (50.0 g) in $q = mc\Delta T$, calculate the energy released per mole of LiCl, in kJ mol^{-1}.

3 marks

c) Jun used the mass of water (50.0 g) in $q = mc\Delta T$. His teacher suggested that using the mass of the whole solution (51.7 g) might be more accurate, because the dissolved LiCl is also heated. Assume the solution has the same specific heat capacity as water. Explain whether using the solution mass would make Jun's calculated value larger or smaller.

2 marks

Question 4 (7 marks)

In a classroom demonstration, solid barium hydroxide octahydrate, $Ba(OH)_2 \cdot 8H_2O$, and solid ammonium chloride, NH_4Cl, are mixed in a small flask. The flask becomes so cold that it can freeze to a wet wooden board.

$$Ba(OH)_2 \cdot 8H_2O(s) + 2NH_4Cl(s) \rightarrow BaCl_2(aq) + 2NH_3(g) + 10H_2O(l)$$

To measure the energy change, a teacher stood the small flask in 75.0 g of water, H_2O, in an insulated container. She added 9.46 g of $Ba(OH)_2 \cdot 8H_2O$ and excess NH_4Cl to the flask. The temperature of the water in the container fell from 21.5 °C to 13.2 °C. Treat the water in the container as the only source of the energy taken in by the reaction.

a) Identify whether this reaction is exothermic or endothermic.

1 mark

b) Calculate the energy, in J, transferred from the water to the reaction.

2 marks

c) Calculate the energy absorbed per mole of $Ba(OH)_2 \cdot 8H_2O$, in kJ mol^{-1}.

3 marks

d) State why the ammonium chloride was added in excess.

1 mark

END OF SECTION B

Additional space

Clearly number any response written here.

END OF TEST

11 Chemistry Unit 2 Topic Test 5: Calorimetry

SECTION A – MULTIPLE-CHOICE ANSWER SHEET

Student name: _____ Teacher: _____

Shade or circle ONE letter for each question. Each question is worth 1 mark. No marks are deducted for incorrect answers.

Question	A	B	C	D
1	(A)	(B)	(C)	(D)
2	(A)	(B)	(C)	(D)
3	(A)	(B)	(C)	(D)
4	(A)	(B)	(C)	(D)
5	(A)	(B)	(C)	(D)
6	(A)	(B)	(C)	(D)
7	(A)	(B)	(C)	(D)
8	(A)	(B)	(C)	(D)
9	(A)	(B)	(C)	(D)
10	(A)	(B)	(C)	(D)

Section A score: _____ / 10